1. Introduction

This study replicates and extends Katovich's (2024) analysis to quantify the causal effects of energy infrastructure on bird populations and species diversity. I employ a difference-in-differences (DiD) framework across 2 different specifications to analyse the effects on bird counts and species diversity.

The primary specification uses a binary treatment indicator (1 = first shale well/turbine constructed within 5km of a CBC circle), isolating energy infrastructure's impact on bird count and biodiversity outcomes. In addition, I explore a continuous measure of treatment intensity to assess marginal effects. We extend the continuous treatment specification by incorporating state-specific linear trends, relaxing the parallel trends assumption. Finally, I examine whether energy infrastructure impacts bird populations indirectly through human population changes and propose an instrumental variable (IV) approach, leveraging refugee resettlement patterns as an extension to isolate exogenous variation in population and strengthen causal claims.

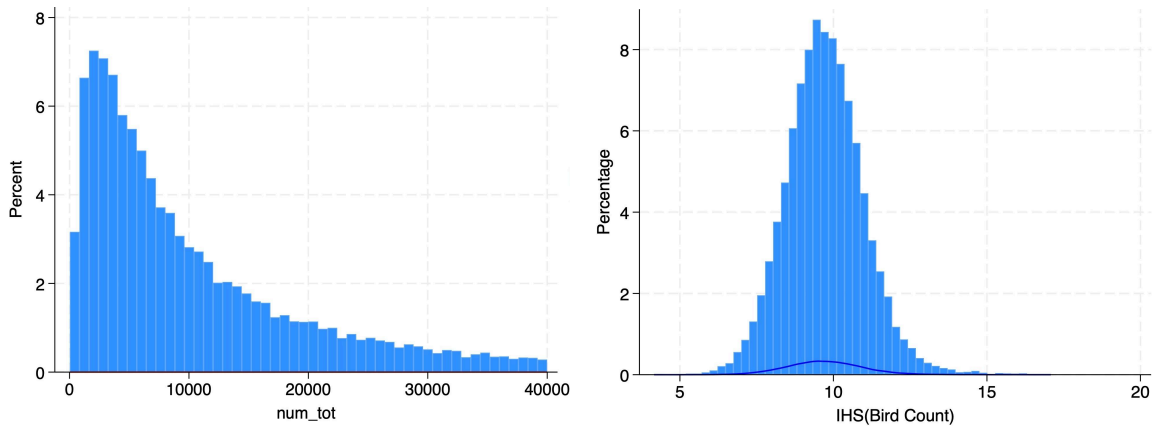
2. Data Treatment

In this study, we employ longitudinal data from the Christmas Bird Count (CBC) spanning 2000 to 2020, encompassing over 31,000 observations of bird counts and species diversity. While CBC surveys occur on a single day in December, limiting their ability to capture annual bird activity, the focus on relative temporal comparisons within CBC circles mitigates seasonal biases.

For variables with very low missingness (< 2–3 %), we elect to drop the affected circle–year observations, since the small loss of data is unlikely to bias our estimates. However, wind-speed and snowfall readings exceed this threshold. To preserve each CBC circle's distinct climate profile, we instead impute these missing values using the circle's own historical mean. Intuitively, this replaces a missing snow or wind-speed observation with the "typical" value for that circle, thereby maintaining the within-circle, over-time variation that identifies our DiD estimates while minimizing sample attrition.

Given the propensity for overcounting errors in the CBC data, stemming from issues such as duplicate counting or misidentification, we winsorize the bird count variables at the 99th percentile. This procedure caps extreme values in the upper tail of the distribution, reducing the risk of attenuation bias that could otherwise drive coefficient estimates towards zero. In contrast, species counts, which generally exhibit lower magnitudes and do not suffer from similar outlier behavior, are left unmodified.

Furthermore, our analysis applies the inverse hyperbolic sine (IHS) transformation to all continuous variables to address the pronounced right skew observed in raw counts, as illustrated in Figures 1 and 2. The IHS transformation compresses the scale for very high values, thereby mitigating the disproportionate influence of extreme observations, while remaining defined at zero without the need for additional constants. As a result, the transformed variables exhibit distributions that closely resemble those derived from a natural logarithm transformation, allowing coefficients to be interpreted as semi-elasticities.



Figures 1 and 2: Pre and post-IHS transformation of the total bird count distribution

Notes: The pre-IHS distribution(Figure 1) has its range restricted to the 90th percentile of the distribution on the upper bound for visualisation. Therefore, the actual right skew is more pronounced than indicated. The post-IHS transformation range has been left unmodified.

A comprehensive set of control variables is utilised, including weather-related conditions, land-use, and counting effort data. County-level human population data is incorporated to test whether energy infrastructure impacts bird populations indirectly through human migration activity.

3. Econometric Strategy

3.1. Binary specification

Treated CBC circles are defined as those where shale wells or wind turbines are constructed either within the geographic boundary of the circle or within a 5km buffer zone. This buffer captures spillover effects, such as habitat fragmentation or noise pollution from nearby infrastructure, which may indirectly impact bird populations within the circle even if the infrastructure lies outside the boundary. We analyze two outcome variables: total bird counts (aggregated across all species) and counts of species within specific ecological groups(e.g., woodland birds, resident birds), both transformed using the IHS function.

Our baseline regression is specified as :

$$y_{ct} = \beta_1 T_{ct} + Z'_{ct}\beta_2 + \gamma_c + \delta_t + \epsilon_{ct}$$

Where y_{ct} represents the IHS-transformed bird count or species diversity in circle *c* and year *t*. We implement a pre/post differences-in-differences specification where we regress the outcome variables on a treatment indicator T_{ct} which is defined as 1 starting in the first year a shale well or wind turbine is constructed within the 5km buffer, and remains 1 in all subsequent years. This specification isolates the sustained ecological effect of infrastructure by establishing clear pre- and post-treatment periods.

To isolate causal effects, we control for a set of variables, Z'_{ct}, which are measured at the circle-year level to mitigate omitted variable bias (OVB) arising from unobserved factors correlated with energy infrastructure placement and bird outcomes. For example, weather conditions, minimum/maximum temperature, wind speed, and snowfall are included because energy projects may cluster in regions with climate patterns that independently affect bird activity. Land use variables (agricultural, pasture, and developed land proportions) account for habitat changes that could simultaneously attract infrastructure development and alter bird populations. Counting effort (number of observers) adjusts for heterogeneity in observer participation across circles and years, ensuring that differences in detection effort do not confound treatment effects.

I also include circle and year fixed effects. Circle fixed effects γ_c capture all the unobserved characteristics of each bird count circle that remain constant over time. Year fixed effects δ_t capture the variation that is common to all bird count circles in a given year *t.* Standard errors are clustered at the circle level to account for potential correlation in the error terms within circles over time.

For our causal interpretation to be credible, two key assumptions must hold. First, the parallel trends assumption requires that, prior to treatment, bird counts and species diversity in treated and untreated circles followed identical trajectories. Second, the conditional independence assumption (CIA) necessitates that, after controlling for time varying confounders, treatment assignment is uncorrelated with unobserved factors affecting the outcome variables. Together, these assumptions ensure that observed post-treatment differences in bird counts are attributable to energy infrastructure, and not omitted variables or pre-existing trends.

3.2 Continuous specification

In the continuous treatment specification, we modify the treatment variable to represent the cumulative number of shale wells or wind turbines operating within a 5km buffer of each CBC circle. This approach captures the marginal ecological impacts of infrastructure intensity. We also apply the IHS transformation to the continuous treatment variable.

The continuous specification imposes stricter assumptions compared to the binary model. First, it assumes a linear relationship between the IHS-transformed treatment variable and outcomes, which may mask nonlinear ecological responses (for example, diminishing returns to infrastructure expansion). The impact on bird populations from an increase in shale wells or wind turbines may not increase proportionally with the installations. Second, the parallel trends assumption requires that, in the absence of treatment, circles with different levels of infrastructure intensity would have followed parallel trends in bird outcomes. This assumption is more difficult to justify under a continuous treatment model, as differing levels of treatment intensity can lead to significant differences in bird populations.

To address these concerns, I extend the model by incorporating state-specific linear trends. This adjustment allows each state to have its own underlying trend in bird counts/species diversity, relaxing

the parallel trends assumption while preserving within-state variation. However, it must be noted that this specification remains subject to the potential for non-linear effects.

4. Results

Effects of Shale Oil and Gas Production

Table 1 presents the estimated impacts of shale well development on bird population counts using both binary and continuous specifications. The arrival of shale wells within 5km of a CBC circle shows no statistically significant effect on total bird counts. However, significant declines are observed for specific groups: Longer Migrants (-18.6%, p = 0.045), Resident Birds (-11.9%, p = 0.021), and Grassland birds (-17.6%, p = 0.006). Since the outcomes were transformed using the inverse hyperbolic sine function, we can interpret semi-elasticities as the number of birds/species counted after the arrival of a shale well/wind turbine near a CBC circle. Semi-elasticities are calculated as follows: $100 \times (e^{(\beta 1)} - 1)$.

The continuous treatment specification estimates elasticities as both outcome and independent variables have been transformed, reflecting marginal effects of infrastructure intensity. Results for the continuous treatment definition shows a statistically significant effect on total bird population counts. Rescaled to a 10% increase in cumulative wells operating, this implies a 0.46% decline in total bird counts (p = 0.017). Proportional declines of similar magnitude are observed across the groups (roughly between -0.3% and -0.6%), all statistically significant at the 5%/10% levels.

Table 1: Results: Effects of Shale Wells on Bird Population Counts

	Total	Grassland	Woodland	Wetland	Other Habitat	Urban	Non-Urban	Residents	Short Migrants	Longer Migrants
Coef. (Binary)	−0.116	−0.194	−0.060	−0.075	−0.076	−0.103	−0.132	−0.127	−0.103	−0.206
St. Error (Binary)	(0.093)	(0.071)	(0.070)	(0.112)	(0.124)	(0.105)	(0.085)	(0.055)	(0.107)	(0.097)
p-value (Binary)	0.215	0.006*	0.396	0.505	0.544	0.326	0.122	0.021*	0.337	0.034*
Coef. (Continuous)	−0.046	−0.046	−0.028	−0.019	−0.051	−0.043	−0.037	−0.034	−0.045	−0.056
St. Error (Continuous)	(0.019)	(0.015)	(0.015)	(0.023)	(0.024)	(0.021)	(0.017)	(0.012)	(0.022)	(0.023)
p-value (Continuous)	0.017*	0.002*	0.063**	0.408	0.036*	0.039*	0.033*	0.003*	0.043*	0.016*
n (Binary)	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079
n (Continuous)	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079

*Note: Numbers in parentheses are cluster-robust standard errors, clustered at the circle level. Continuous outcomes and treatment are transformed using inverse hyperbolic sine function. * and ** denotes significance at the 5% and 10% level, respectively.*

Table 2 reports the results for the effect of shale well arrival and the number of shale wells operating on Bird Species Count. It does not have a statistically significant effect on total species counts, and only weakly reduces species diversity among residents by 2.76% (p = 0.054). The continuous treatment also does not show a measurable effect, but does reduce Woodland species diversity by 0.07% (p = 0.090) and Residents species diversity by 0.1% (p = 0.000) for a 10% increase in the number of shale wells operating.

Table 2: Results: Effects of Shale Wells on Bird Species Count

	Total	Grassland	Woodland	Wetland	Other Habitat	Urban	Non-Urban	Residents	Short Migrants	Longer Migrants
Coef. (Binary)	–0.014	–0.037	–0.002	0.017	0.010	–0.003	–0.025	–0.028	0.001	–0.058
St. Error (Binary)	(0.021)	(0.029)	(0.017)	(0.049)	(0.020)	(0.016)	(0.031)	(0.014)	(0.023)	(0.041)
p-value (Binary)	0.494	0.204	0.910	0.729	0.623	0.861	0.428	0.054**	0.982	0.155
Coef. (Continuous)	–0.006	–0.009	–0.007	0.001	–0.001	–0.004	–0.008	–0.010	–0.003	–0.014
St. Error (Continuous)	(0.004)	(0.006)	(0.004)	(0.009)	(0.004)	(0.004)	(0.006)	(0.003)	(0.005)	(0.009)
p-value (Continuous)	0.144	0.139	0.090**	0.954	0.821	0.240	0.193	0.000*	0.547	0.107
n (Binary)	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079
n (Continuous)	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079

*Note: Numbers in parentheses are cluster-robust standard errors, clustered at the circle level. Continuous outcomes and treatment are transformed using inverse hyperbolic sine function. * and ** denotes significance at the 5% and 10% level, respectively.*

Effects of Wind Energy Installation

Table 3 reports the results of the binary and continuous treatment on Bird Population Count, with both showing statistically insignificant effects on total bird counts. The binary treatment shows an increase in Woodland bird counts by 8.4% (p = 0.001), indicating potential localized habitat benefits. However, the continuous treatment yields no significant effects, implying that the effects may not scale linearly with the number of turbines operating.

Table 3: Results: Effects of Wind Turbines on Bird Population Count

	Total	Grassland	Woodland	Wetland	Other Habitat	Urban	Non-Urban	Residents	Short Migrants	Longer Migrants
Coef. (Binary)	0.008	0.036	0.081	0.023	–0.011	0.007	0.026	0.021	–0.001	0.001
St. Error (Binary)	(0.033)	(0.031)	(0.024)	(0.057)	(0.040)	(0.034)	(0.038)	(0.024)	(0.034)	(0.052)
p-value (Binary)	0.809	0.243	0.001*	0.679	0.802	0.841	0.494	0.384	0.969	0.981
Coef. (Continuous)	0.005	0.016	0.013	0.009	0.006	0.004	0.012	0.003	0.002	0.014
St. Error (Continuous)	(0.012)	(0.010)	(0.008)	(0.017)	(0.016)	(0.013)	(0.012)	(0.008)	(0.012)	(0.016)
p-value (Continuous)	0.677	0.116	0.103	0.560	0.693	0.767	0.302	0.672	0.863	0.380
n (Binary)	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079
n (Continuous)	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079

*Note: Numbers in parentheses are cluster-robust standard errors, clustered at the circle level. Continuous outcomes and treatment are transformed using inverse hyperbolic sine function. * and ** denotes significance at the 5% and 10% level, respectively.*

Table 4 shows the effect of Wind Turbines on Bird Species diversity. Again, both treatments show statistically insignificant results on total bird species diversity. Binary treatment effects emerge for Woodland (1.71% , p = 0.026) and Resident (1.3%, p = 0.054) species, while continuous estimates remain insignificant. This may suggest that wind turbine impacts may be spatially concentrated which this treatment is not suited to detect.

Table 4: Results: Effects of Wind Turbines on Bird Species Count

	Total	Grassland	Woodland	Wetland	Other Habitat	Urban	Non-Urban	Residents	Short Migrants	Longer Migrants
Coef. (Binary)	0.009	0.012	0.017	–0.003	0.001	0.009	0.008	0.013	0.007	0.009
St. Error (Binary)	(0.008)	(0.012)	(0.008)	(0.019)	(0.009)	(0.006)	(0.012)	(0.007)	(0.009)	(0.018)
p-value (Binary)	0.225	0.323	0.026*	0.885	0.940	0.121	0.520	0.053**	0.429	0.596
Coef. (Continuous)	0.002	0.006	0.003	0.001	–0.000	0.002	0.003	0.003	0.002	0.004
St. Error (Continuous)	(0.003)	(0.004)	(0.003)	(0.006)	(0.003)	(0.002)	(0.004)	(0.002)	(0.003)	(0.005)
p-value (Continuous)	0.405	0.123	0.310	0.909	0.946	0.330	0.538	0.153	0.584	0.401
n (Binary)	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079
n (Continuous)	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079	30,079

*Note: Numbers in parentheses are cluster-robust standard errors, clustered at the circle level. Continuous outcomes and treatment are transformed using inverse hyperbolic sine function. * and ** denotes significance at the 5% and 10% level, respectively.*

5. Extension: State-Specific Linear Trends

Table 5: Continuous DiD with State–Specific Linear Trends

	Total	Grassland	Woodland	Wetland	Other Habitat	Urban	Non-Urban	Residents	Short Migrants	Longer Migrants
Wells on Bird Population Counts										
Coef. (Cont.)	−0.045	−0.041	−0.024	−0.006	−0.052	−0.041	−0.038	−0.026	−0.045	−0.050
St. Error	(0.019)	(0.015)	(0.014)	(0.022)	(0.023)	(0.020)	(0.017)	(0.011)	(0.021)	(0.023)
p-value	0.014*	0.005*	0.101	0.788	0.024*	0.042**	0.029*	0.021*	0.033*	0.029*
Wells on Bird Species Counts										
Coef. (Cont.)	−0.005	−0.008	−0.005	0.001	−0.003	−0.002	−0.009	−0.007	−0.004	−0.010
St. Error	(0.004)	(0.006)	(0.004)	(0.009)	(0.004)	(0.003)	(0.006)	(0.003)	(0.005)	(0.008)
p-value	0.184	0.195	0.254	0.943	0.405	0.475	0.132	0.020*	0.442	0.210
Turbines on Bird Population Counts										
Coef. (Cont.)	0.002	0.019	0.009	0.004	0.005	0.000	0.011	0.007	−0.001	0.022
St. Error	(0.012)	(0.010)	(0.008)	(0.017)	(0.016)	(0.013)	(0.012)	(0.008)	(0.012)	(0.016)
p-value	0.857	0.071**	0.249	0.820	0.744	0.994	0.338	0.430	0.914	0.165
Turbines on Bird Species Counts										
Coef. (Cont.)	0.002	0.005	0.001	0.000	0.000	0.001	0.002	0.002	0.001	0.005
St. Error	(0.003)	(0.004)	(0.003)	(0.006)	(0.003)	(0.002)	(0.004)	(0.002)	(0.003)	(0.005)
p-value	0.565	0.212	0.584	0.973	0.882	0.588	0.613	0.350	0.753	0.288

*Note: Numbers in parentheses are cluster-robust standard errors, clustered at the circle level. Continuous outcomes and treatment are transformed using inverse hyperbolic sine function. * and ** denotes significance at the 5% and 10% level, respectively.*

As detailed previously, the DiD method hinges on the parallel trends assumption, which assumes that treatment and control groups follow a common trend in the absence of treatment. However, with the continuous specification involving many states and years with varying levels of infrastructure intensity, this assumption becomes harder to justify. To address this issue, I relax the parallel trends assumption by introducing state-specific linear trends - by including interactions between state dummies and year dummies. This model assumes that in the absence of the treatment effect, bird counts/species diversity in state *s* deviates from common trends by following a linear trend captured by θ_s. The results are close to the baseline specification which supports the use of the DiD approach.

$$y_{ct} = \beta_1 T_{ct} + Z_{ct}^{'}\beta_2 + \gamma_c + \delta_t + \sum_{s=1}^{S} \theta_s (State_s \cdot t) + \epsilon_{ct}$$

In terms of differences with the baseline, there is a minimal impact on the coefficients for shale wells. The p-values are slightly larger, suggesting a reduction in precision due to the additional variability captured by state-specific trends. For species diversity, only one species remains statistically significant, compared to two in the baseline.

For wind turbines, one new statistically significant result appeared for bird counts, which wasn't seen in the baseline model. Despite this, coefficients remain close to zero, which may indicate localized effects that this model isn't designed to capture. No significant results appear for turbines on species diversity, with coefficients similar to the baseline *(See Table 5 for the results)*.

6. Changes in Human Population

To investigate whether energy infrastructure impacts bird populations indirectly through human migration activity, independent of direct land use impacts, I first estimate the effect of shale well and wind turbine arrival on county-level human population. We run the following DiD framework:

$$IHS(Population_{ct}) = \beta_1 T_{ct} + X_{ct}\gamma + \mu_c + \delta_t + \epsilon_{ct}$$

Where T_{ct} represents the well/turbine arrival indicator, X_{ct} is the control vector(controlling for temperature, windspeed, snowfall and counting effort), μ_c and δ_t represents circle and year fixed effects, respectively. I then regress the IHS-transformed total-bird count on human population using a continuous difference-in-differences specification.

Table 5 shows that shale-well arrival reduces county-level population by 3.4% (p = 0.003), while turbine arrival lowers it by 1.6% (p = 0.009). Crucially, increases in human population have no statistically significant effect on the total number of birds counted, yielding a coefficient of -0.059 (p = 0.557). Hence, the energy infrastructure impact on birds does not appear to be mediated by human resettlement.

Table 6: Human Population Analysis

Effect of Shale Well Arrival on Human Pop.	
Coef (CS)	−0.035
St. Error	(0.012)
p-val	0.003
n	30,079
Effect of Wind Turbine Arrival on Human Pop.	
Coef (CS)	−0.016
St. Error	(0.006)
p-val	0.009
n	30,079
Effect of Human Pop. on Birds Reported	
Coef (DID)	−0.059
St. Error	(0.100)
p-val	0.557
n	30,079

Note: Numbers in parentheses are cluster-robust standard errors, clustered at the circle level. Continuous outcomes and treatment are transformed using inverse hyperbolic sine function.

6.1. Instrumental Variable Approach: Refugee Resettlement Programs

To isolate the causal effect of human population on bird counts, I propose using a migration shock, such as refugee resettlement patterns under the US Refugee Admissions Program (USRAP)[1] as an

[1] U.S Citizenship and Immigration Services. (n.d.). *U.S. Refugee Admissions Program (USRAP).* U.S. Department of Homeland Security.

instrumental variable (IV) for population levels in a two-stage least squares (2SLS) framework. Refugee inflows have been widely used as exogenous shocks to local populations in empirical studies (e.g., Card 1990; Angrist & Kugler 2003), as placement decisions are driven by federal policy rather than local environmental factors. This makes refugee arrival a plausible source of exogenous variation.

The instrument could be constructed using publicly available data from the Office of Refugee Resettlement[2] (ORR) which report annual refugee arrivals at the county level from 1975-2018. I instrument county-year human population using the number of refugees initially resettled in that county under USRAP per capita, lagged by one year. This scaling accounts for differences in county size and preserves consistency with my treatment of continuous variables in this project. Moreover, dividing the refugee inflow by the prior year's population ensures the instrument reflects an addition to a predetermined stock of residents and preserves the necessary temporal ordering, namely, that the instrument is known before the outcome; thereby guarding against mechanical collinearity and reverse causality (Angrist & Pischke, 2009).

The first-stage is specified as follows:

$$IHS(Population_{ct}) = \alpha_0 + \alpha_1(IHS(\frac{Refugees_{ct}}{Pop_{c,t-1}})) + \alpha_2 X_{ct} + \mu_c + \delta_t + \epsilon_{ct}$$

In the second stage, I use the predicted values from the first stage to estimate the effects on bird counts:

$$IHS(Bird\ Count_{ct}) = \beta_0 + \beta_1(IHS(\widehat{Population}_{ct})) + \beta_2 X_{ct} + \eta_c + \gamma_t + \varepsilon_{ct}$$

This IV strategy addresses potential reverse causality and endogeneity. For example, reverse causality could arise if areas with declining bird populations experience migration or population reductions due to local environmental degradation, which could confound the relationship between population and bird counts.

For the IV to be valid, three conditions must hold. First, refugee arrival must have a causal effect on county-level population. Second, the exclusion restriction requires that refugee resettlement affects bird counts solely through human population changes. Third, exogeneity demands that refugee placement is uncorrelated with unobserved confounders, such as pre-existing environmental degradation.

There exist limitations with this approach. There could be potential weakness with the instrument in counties with few refugees, attenuation due to secondary migration (post-arrival) and the possibility that settlement locations do not coincide with infrastructure impacts. This approach offers a feasible extension given that I couldn't identify a credible source of exogenous population shocks in the primary dataset. Although turbine and well installations are linked to modest population declines, limiting the

[2] "Dreher, A., Langlotz, S., Matzat, J., Parsons, C. and Mayda, A. (2020). Immigration, Political Ideologies and the Polarization of American Politics. CEPR Discussion Paper 15587."

scope for a population-mediated effect on bird counts; implementing this IV extension nonetheless adds completeness by formally testing and ruling out the indirect pathway.

7. Conclusion

In this study, I replicate and extend the results from Katovich (2024), confirming significant negative impacts of shale wells on bird counts. However, the same level of statistical significance is not observed for species diversity and wind turbines.

To extend these results, I estimate a state-specific linear trends model to explore the potential impact of state-level confounders. In examining potential mediation effects through human population, we propose an IV approach, leveraging refugee resettlement patterns to isolate exogenous population variation, though data limitations preclude implementation here. Future extensions could explore spatially localized impacts or non-linear responses to refine causal mechanisms.

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